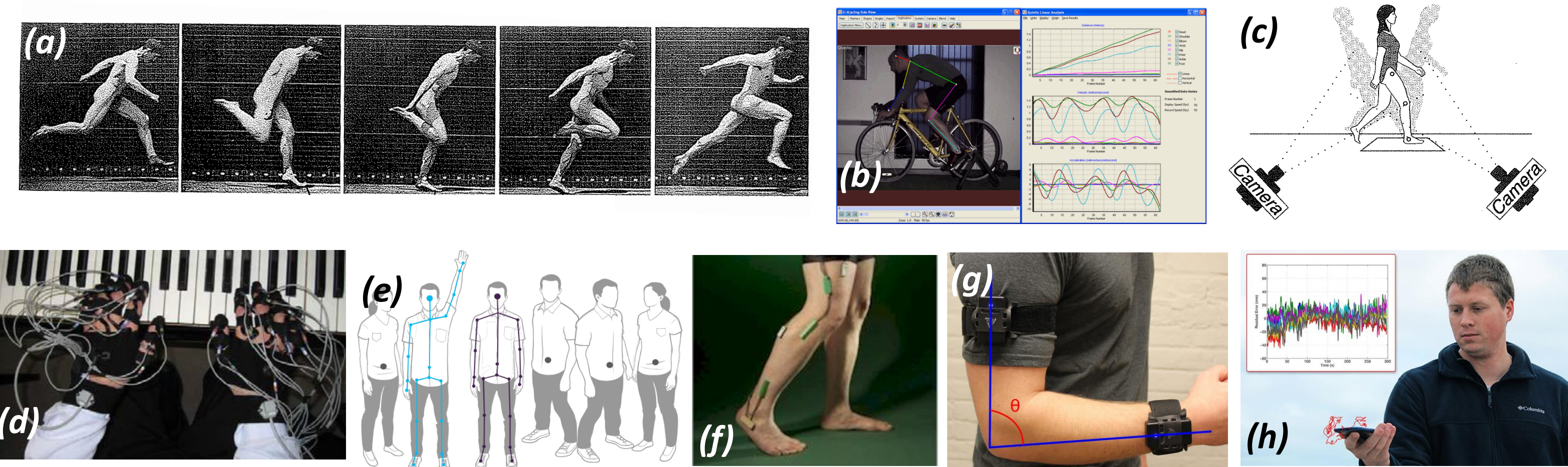


Engineering Design of a Hybrid 3D Motion Tracking System

Eric G. Meyer PhD, Biomedical Engineering, Lawrence Technological University, Southfield, MI

Background and Capabilities

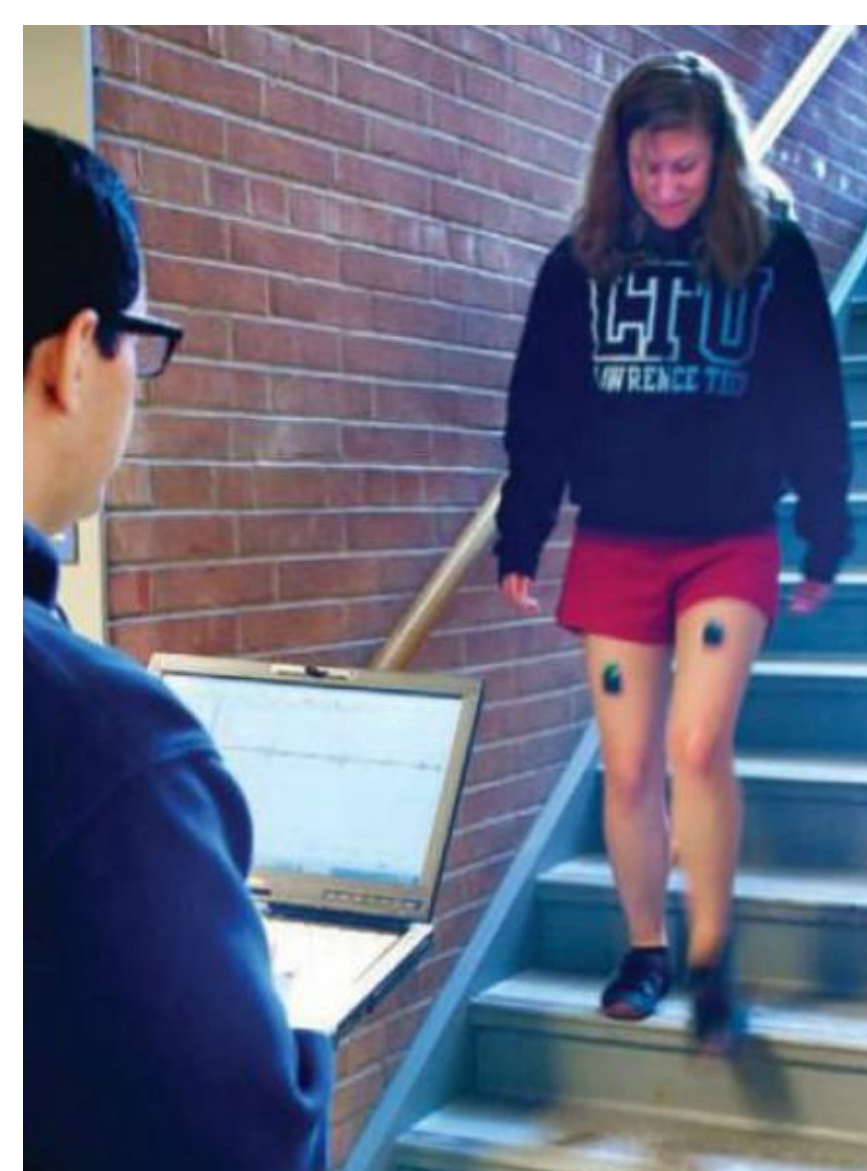
- Biomechanics Laboratories have used many technologies to experimentally measure human kinematics, including; Stroboscopic photography (a), Cinematography (b), Motion capture using Optical (c), or Electromagnetic (d) sensors and markerless tracking (e), Electrogoniometry (f), Inertial sensors (g) Barometry and Global Positioning (h).



- LTU's Experimental Biomechanics Lab (EBL) has capabilities to collect data with a number of these techniques for teaching research, and computer applications.



10 Camera Vicon system measures 3D position and ground reaction forces



16 wireless Delsys sensors measure 3D acceleration and muscle activity



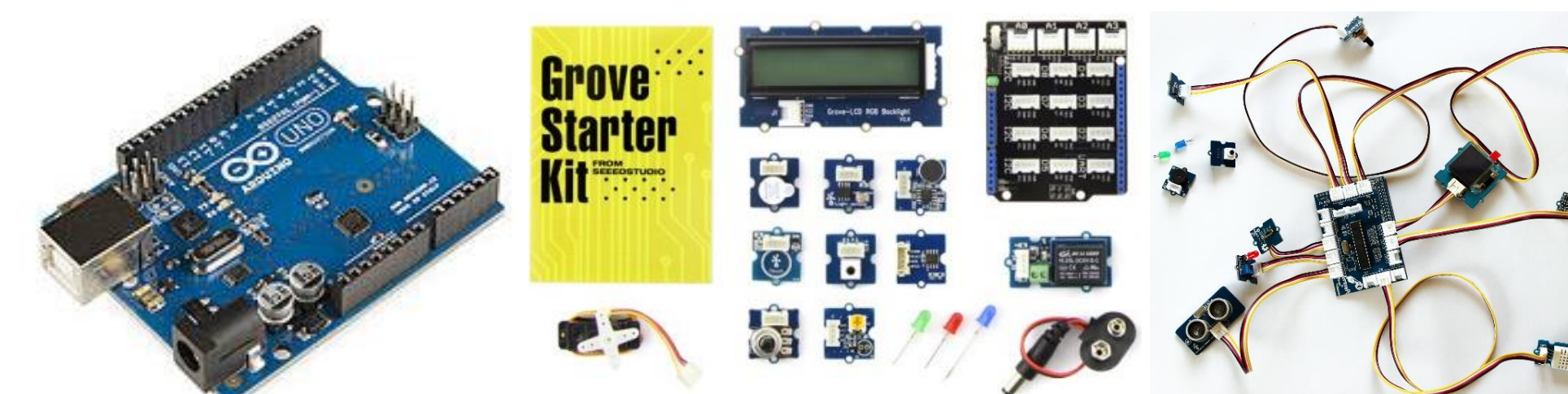
16 wireless PrioVR sensors measure 3D accelerations and orientations

Objective

- Develop a hybrid video and inertial sensor system that uses data fusion to overcome common problems and errors typically associated with either type of motion tracking

Design and Prototypes

- Open computer hardware systems like Arduino and BeagleBone with plug-and-play electronic sensors like SEEED Grove or Adafruit were investigated to build a custom inertial sensing system.



Initial prototype testing: Arduino and Grove IMU



Final Design: BeagleBone and Adafruit absolute IMU

BeagleBone Black

- 1 GHz ARM Cortex-A8
- 512 MB DDR3 RAM
- 2 GB Storage
- WiFi 802.11 b/g/n
- Bluetooth 4.1 LE
- USB, analog and digital ports

Adafruit BNO055

- Acceleration: Three axis of acceleration (gravity + linear motion) in m/s^2 .
- Angular Velocity: Three axis of 'rotation speed' in rad/s.
- Absolute Orientation: (Euler Angles) Three axis orientation data based on a 360° sphere.

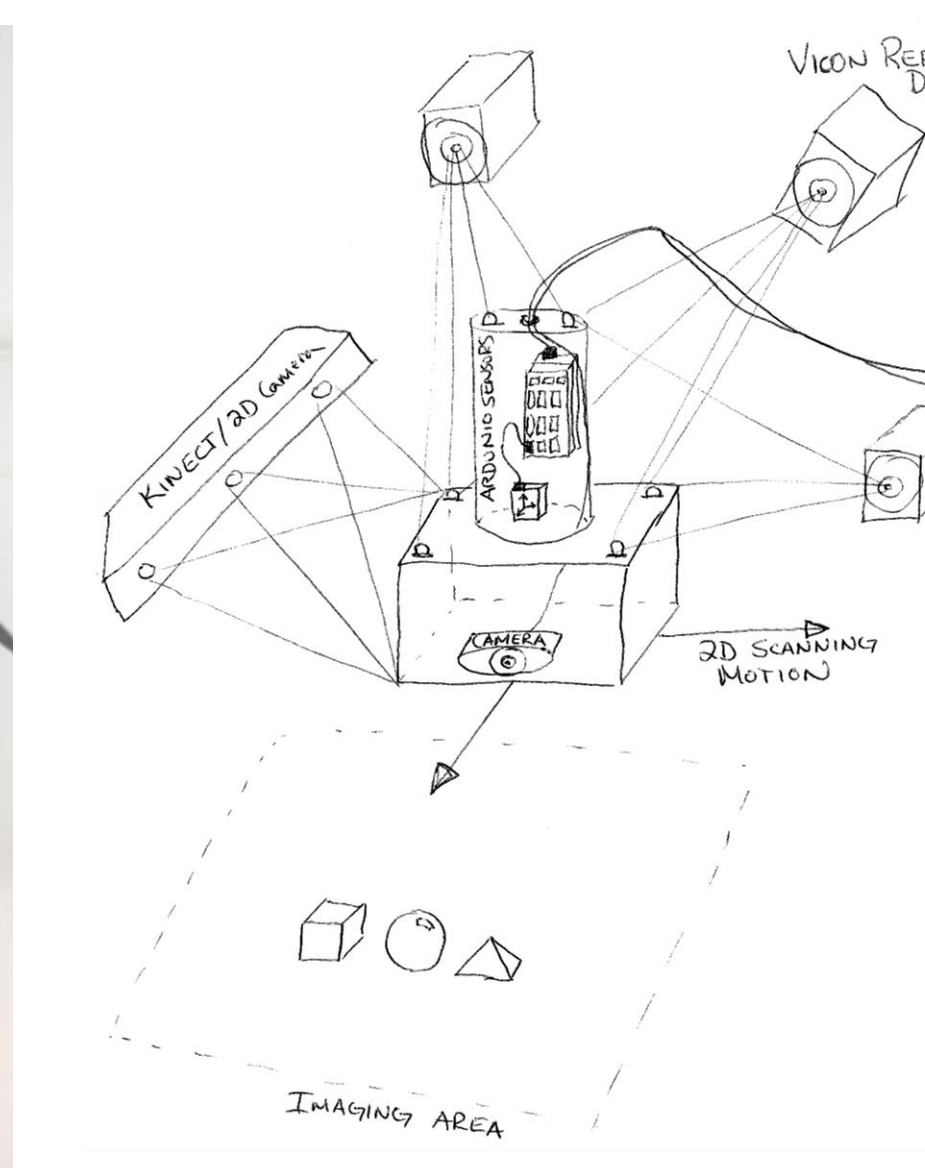
- A 2D computer vision program and the Microsoft Kinect software development kit were considered for the video based position tracking component.



Computer vision systems: Kinect or USB HD video camera



Prototype validation testing: Video data of scanning area matched with position data.



Vicon motion capture used to validate position

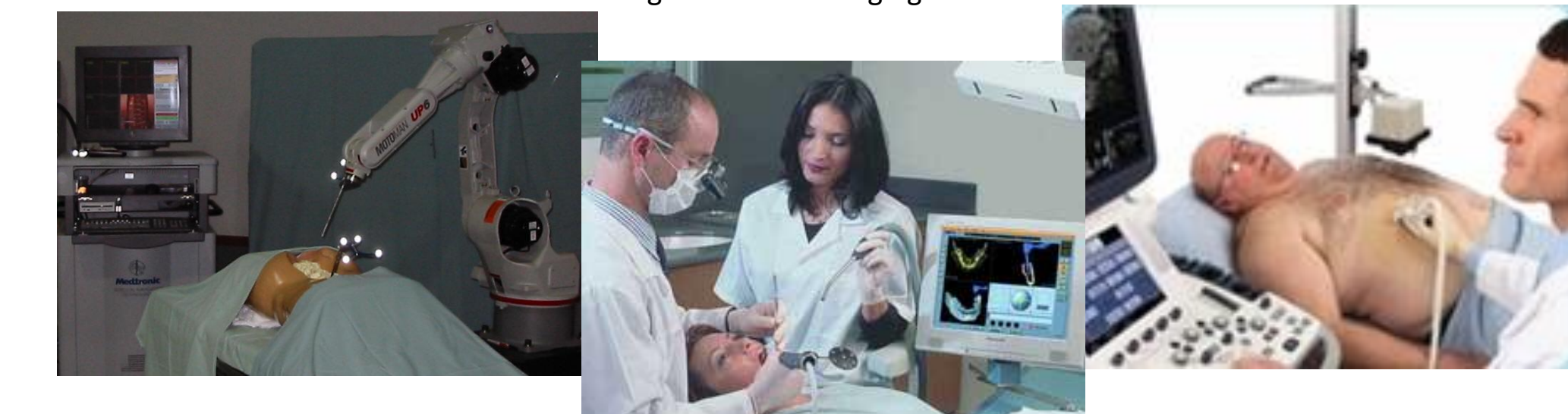
Applications and Future Work

- There are many applications for real time collection and streaming of rigid body motion data.

"Quantified Self" (QS) consumer products like smartphones, activity monitors, video game and virtual reality systems.



Medical devices for robotic or computer navigated surgery, patient monitoring and 3D scanning ultrasound imaging.



- The size, weight and cost of inertial sensing transducers and video cameras has dropped dramatically. However, errors in inertial sensors during slow movements, and lag in video data for high speed movements or when the line of sight is impeded may limit their accuracy and application as standalone solutions.

References

- Erickson BJ, Jack CR. Correlation of Single Photon Emission CT with MR Image Data Using Fiducial Markers. Am J Neuroradiology, 14, 713-720, 1993.
- Prager RW, Rohling RN, Gee AH, Berman L. Rapid Calibration For 3-D Freehand Ultrasound. Ultrasound Med Bio. 24(6), 855-869, 1998.
- Böniga J, Fischera C, Weckenda H, Döbereiner F, Franke J. Accuracy and immersion improvement of hybrid motion capture based real time virtual validation. Proc CIRP, 21, 294-299, 2014.
- Chen C, Jafari R, Kehtarnavaz N. Fusion of Depth, Skeletal and Inertial Data for Human Action Recognition. ICASSP, 2712-2716, 2016.